

## RESEARCH ARTICLE

# Enhancing Vision Transformers for Facial Expression Recognition

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**ABSTRACT** Facial expression recognition (FER) remains a challenging problem due to subtle intra-class variations, strong inter-class similarities, and external factors including pose variations, illumination changes, and occlusions. While vision transformers (ViTs) have demonstrated success in general image classification, their ability to model both global semantic context and localized discriminative facial features remains suboptimal for FER. The enhanced vision transformer architecture for facial expression recognition in this study introduces two important innovations. We first create a two-stream representation that simultaneously employs the class token and global average pooling (GAP) to guarantee a more thorough representation that preserves sensitive spatial details while capturing the overall facial context. Second, we present enhanced conditional positional encoding (ECPE), a sophisticated positional encoding technique that captures both local and global spatial dependencies to improve feature learning, and weighted global average pooling (WGAP), a spatially aware pooling mechanism that dynamically emphasizes discriminative facial regions. We evaluate our framework using three prominent FER benchmarks: FER2013, RAF-DB, and FERPlus, exhibiting consistent performance enhancements over conventional transformer baselines and exceeding many state-of-the-art approaches. Ablation studies demonstrate that each element—two-stream representation, WGAP, and ECPE—independently enhances feature learning and classification accuracy. Our approach improves the capacity of vision transformers to recognize subtle facial expressions by integrating global and local feature learning. The approach we present, although specifically tailored for FER, can also be adapted for other fine-grained classification problems necessitating precise discrimination.

**INDEX TERMS** Deep learning, facial expression recognition, vision transformers.

## I. INTRODUCTION

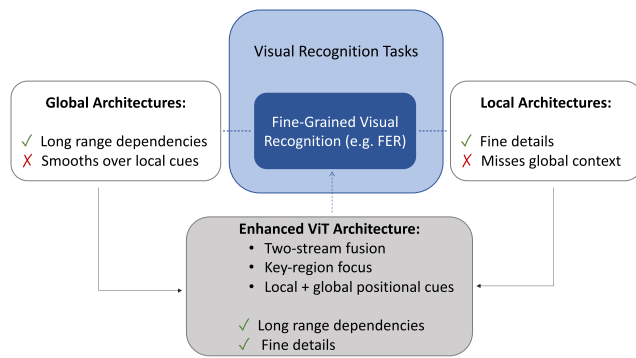
Facial expression recognition (FER) is a core area in computer vision, with diverse practical uses spanning surveillance, medical and psychological diagnostics, human-computer interactions, and robotics [22]. Despite significant progress from deep learning approaches, accurately distinguishing subtle variations among facial expressions remains difficult.

Facial expression recognition (FER) techniques have primarily depended on convolutional neural networks (CNNs). Convolutional Neural Networks (CNNs) are efficient at extracting local features yet often encounter difficulties in accurately modeling global semantic contexts and long-range spatial relationships. Recent studies have attempted to

address these limitations by incorporating attention mechanisms into CNN architectures [7], [28], [40], thereby improving their feature-selection abilities. Nevertheless, CNN-based methods inherently struggle to capture subtle yet essential spatial relationships, which are critical for accurately recognizing subtle facial expressions.

The emergence of Vision Transformers (ViTs) [5], leveraging self-attention mechanisms, has significantly shifted the landscape of visual recognition. ViTs have shown impressive results due to their ability to simultaneously consider local details and global context. However, current transformer-based FER models frequently fail to effectively fuse global context with detailed spatial features necessary for fine-grained FER tasks. Specifically, standard transformer architectures typically use a class token primarily for aggregating global information, potentially neglecting fine-grained spatial features critical for subtle expression differentiation.

The associate editor coordinating the review of this manuscript and approving it for publication was Szidonia Lefkovits<sup>1</sup>.



**FIGURE 1.** Global/Local architecture shortcomings in FER and our enhanced ViT solution.

To address these issues, we propose an optimized vision transformer framework specifically designed for facial expression recognition. Our methodology presents two primary improvements. Initially, we utilize a two-stream representation that combines the class token with global average pooling (GAP), thereby encompassing both global context and essential local spatial features. Secondly, we present weighted global average pooling (WGAP) and enhanced conditional positional encoding (ECPE). WGAP adaptively emphasizes significant facial areas, including the eyes and mouth, allowing the model to underscore distinguishing characteristics. ECPE augments the conventional conditional positional encoding (CPE) [3] by incorporating both local and global spatial dependencies, thereby improving the robustness of feature extraction.

In summary, our main contributions are:

- **Two-Stream Representation:** Combining class token and GAP to capture both global and local features.
- **Weighted Global Average Pooling:** A learnable pooling mechanism that emphasizes key facial features.
- **Enhanced Conditional Positional Encoding:** Extending CPE to incorporate both local and global positional cues.

We validate our proposed framework through extensive experiments on three widely recognized datasets: FER2013 [11], FERPlus [2], and RAF-DB [23]. Our detailed ablation studies systematically evaluate WGAP and ECPE's individual impacts, benchmarking our method against recent transformer-based approaches and state-of-the-art FER models.

Although our research primarily focuses on FER, the methodology established here is widely applicable to other fine-grained classification tasks necessitating accurate differentiation of subtle variations. Our approach significantly enhances visual recognition frameworks by effectively integrating global and local spatial features.

The remainder of the paper is organized as follows: Section II reviews related studies on vision transformers and current facial expression recognition methodologies. Section III outlines the proposed methodology, including our two-stream approach, WGAP, and ECPE. Section IV presents

experimental evaluations and analyses. At last, Section V summarizes principal findings and outlines possibilities for future study.

## II. RELATED WORK

### A. VISION TRANSFORMERS FOR VISUAL RECOGNITION

The transformer architecture, initially proposed by Vaswani et al. [38] for natural language processing, has shown considerable potential in computer vision. Vision transformers (ViTs) utilize self-attention mechanisms to capture long-range dependencies and contextual relationships, surpassing conventional CNNs on multiple benchmarks. Dosovitskiy et al. [5] demonstrated the effectiveness of transformers in image classification, prompting more research into their application across diverse visual tasks.

Recent studies have suggested enhancements to standard transformers to improve computational efficiency and hierarchical feature learning. Liu et al. [25] presented the Swin Transformer, which utilizes local window-based attention to achieve scalable performance on high-resolution images. Pyramid-based architectures, including PVT [41], HVT [32], and PIT [12], utilize hierarchical feature extraction that allow transformers to effectively model both local and global representations. Alternative hybrid approaches, such as CPVT [3] and CVT [43], combine convolutional layers with transformers to improve efficiency and feature learning.

Twins [4] introduced spatially separable self-attention mechanisms that enhance the extraction of local and global features. Conditional Positional Encoding (CPE) [3] introduced an adaptive positional encoding mechanism that dynamically incorporates local spatial relationships, demonstrating its efficacy for tasks requiring spatially-aware representations. Our research extends these advancements by proposing an optimized transformer framework that adeptly integrates global context with localized fine-grained details for facial expression recognition.

### B. FACIAL EXPRESSION RECOGNITION METHODS

Facial expression recognition (FER) has been thoroughly studied using deep learning models, such as convolutional neural networks (CNNs), attention mechanisms, and, more recently, vision transformers. Conventional CNN-based methods extract hierarchical features from facial images but frequently face difficulties in detecting subtle variations in expressions due to limited global receptive fields.

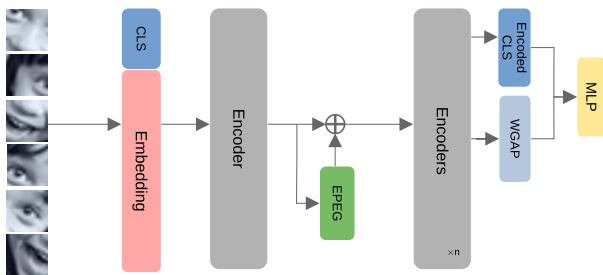
Earlier CNN-based FER models employed conventional architectures like AlexNet [20] and VGGNet [36]. Yu and Zhang [47] proposed an ensemble of CNNs optimized using hinge loss weighting for the purpose of expression recognition. Yao et al. [46] enhanced recognition performance by incorporating residual connections in CNNs. Zhao et al. [49] investigated feature selection methods in AlexNet to derive more discriminative features. Wu et al. [44] used cross-hierarchy contrastive learning to regularize CNN feature extraction and fuse them for robust global FER

representations. Recent CNN-based methods have attempted to improve facial expression recognition (FER) via self-supervised learning [21], emotional education mechanisms [17]. Nevertheless, CNNs naturally fail to capture long-range dependencies, making them suboptimal for subtle tasks such as FER.

Attention-based FER methods have been implemented to tackle these challenges. Fan et al. [7] introduced an attention mechanism highlighting salient facial regions, including the eyes and mouth, essential for expression recognition. Wang et al. [40] presented the Self-Cure Network, which improves facial expression recognition by reducing label noise via adaptive weighting of ambiguous samples and a systematic relabeling procedure to enhance annotation reliability. Huang et al. [13] introduced grid-wise attention in early CNN layers and transformer attention on high-level semantic tokens. Pecoraro et al. [34] presented channel-wise self-attention, improving the model's capacity to focus on expression-relevant features while eliminating background noise.

Researchers have adapted transformer architectures for facial expression recognition (FER) with the advent of vision transformers, utilizing self-attention to improve expression recognition. Xue et al. [45] introduced a multi-head self-attention dropout mechanism to enhance part-based feature modeling. Ma et al. [27] proposed a feature fusion strategy that enhances attention distribution, resulting in superior performance on conventional FER benchmarks. Kim et al. [18] devised a transformer-based model featuring an improved attention mechanism, achieving commendable outcomes on challenging FER datasets. Aouayeb et al. [1] studied a vision transformer trained in alongside squeeze-and-excitation blocks to enhance feature representation.

Despite significant advancements in transformer-based FER models, challenges persist in accurately modeling subtle expression cues due to restricted spatial sensitivity and generic positional encodings. To address these limitations, we propose specific architectural improvements as outlined in the subsequent section.



**FIGURE 2.** General overview of the proposed transformer-based architecture for facial expression recognition.

### III. PROPOSED METHOD

Facial expression recognition requires the ability to spot subtle differences in facial features while maintaining resilience against variations in pose, occlusions, and identity biases.

Although vision transformers proficiently capture long-range dependencies, they frequently struggle to achieve an optimal equilibrium between global semantic comprehension and localized discriminative features, which are essential for facial expression recognition. To tackle these challenges, we propose a transformer-based framework specifically designed for facial expression recognition, incorporating a two-stream representation and task-specific improvements, including weighted global average pooling (WGAP) and enhanced conditional positional encoding (ECPE).

Figure 3 demonstrates the comprehensive architecture of our model. The framework is based on a vision transformer backbone that employs conditional positional encoding (CPE) [3], incorporating the subsequent principal innovations. The symbols and operators used throughout this section are summarized in Table 1.

**TABLE 1.** Notation used in the proposed enhanced vision transformer framework.

| Symbol/Operator             | Description                                 |
|-----------------------------|---------------------------------------------|
| $\mathbf{x}$                | Input image                                 |
| $\mathbf{z}_0^m$            | input of first TE at stage $m$              |
| $\mathbf{y}_0^m$            | Output of first TE at stage $m$             |
| $\mathbf{y}_1^m$            | TE output after CPE addition at stage $m$   |
| $\mathbf{y}_n^m$            | Output of the $n$ -th TE block at stage $m$ |
| $\mathbf{y}_N^M$            | Output of the last TE block at last stage   |
| $\mathbf{y}_{\text{class}}$ | Extracted Class token                       |
| $\mathbf{y}_{\text{WGAP}}$  | Weighted global average pooled feature      |
| $\mathbf{y}_{\text{final}}$ | Final class logits                          |
| $\text{PE}(\cdot)$          | Patch Embedding                             |
| $\text{TE}(\cdot)$          | Transformer Encoder block                   |
| $\text{CPE}(\cdot)$         | Positional Encoding Generator               |
| $\text{ECPE}(\cdot)$        | Enhanced Positional Encoding Generator      |
| $\text{WGAP}(\cdot)$        | Weighted Global Average Pooling             |

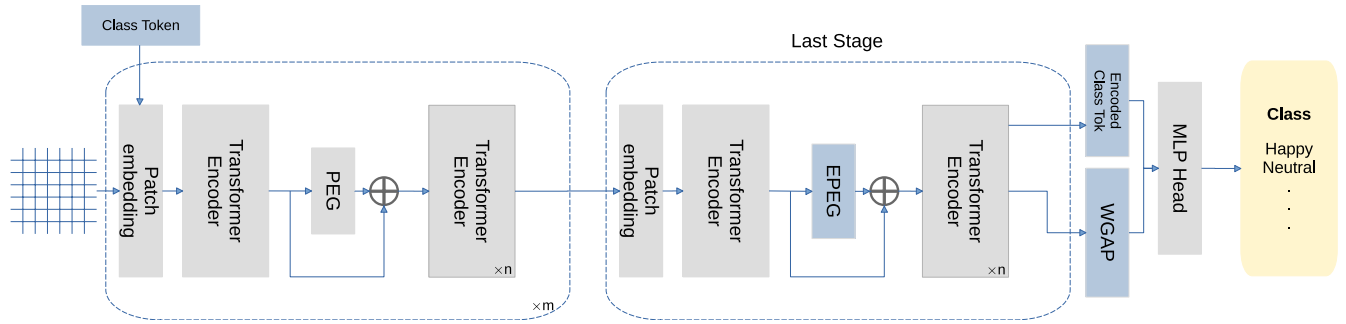
#### A. TWO-STREAM REPRESENTATION: CLASS TOKEN AND GLOBAL AVERAGE POOLING

In conventional vision transformers, the class token aggregates information from all spatial patches and functions as the global feature for classification. Although effective for general tasks, exclusive reliance on the class token often overlooks vital localized facial cues, including subtle changes in mouth curvature, eye wrinkles, or eyebrow movement, which are important to facial expression recognition.

To address this limitation, we present a two-stream representation that integrates:

- The class token (CT), which captures high-level semantic information through self-attention.
- Global average pooling (GAP) applies on the spatial patch embeddings in the final transformer layer to aggregate localized discriminative features.

Given an input image, the transformer processes it into a sequence of feature tokens, containing spatial tokens and a specific class token. Rather than depending exclusively on the class token for classification, we formulate the final representation by concatenating the class token output with the global average pooled spatial features. This integration



**FIGURE 3.** Overview of the proposed architecture. The model integrates class token and global average pooling in a two-stream architecture and incorporates weighted global average pooling (WGAP) and enhanced conditional positional encoding (ECPE) for facial expression recognition.

enables the preservation of both identity-invariant global context and fine-grained discriminative details, which is vital to accurately capturing subtle facial expressions.

### B. WEIGHTED GLOBAL AVERAGE POOLING (WGAP)

Conventional global average pooling (GAP) assigns equal importance to all spatial regions, which is inadequate for facial expression recognition (FER) since certain facial areas, such as the eyes and mouth, play a more significant role in expression identification than others. We propose weighted global average pooling (WGAP) to dynamically learn spatial significance and emphasize critical areas.

As mentioned in Table 1,  $\mathbf{y}_N^M \in \mathbb{R}^{H_M \times W_M \times D}$  denote the output of the last Transformer block of the  $M$ -th Stage, with  $H_M$ ,  $W_M$ , and  $D$  representing the height, width, and channel number respectively. The weighted pooling operation is expressed as follows:

$$\mathbf{y}_{WGAP} = \sum_{i=1}^{H_M} \sum_{j=1}^{W_M} w_{ij} \mathbf{y}_N^M(i, j) \in \mathbb{R}^{1 \times D}. \quad (1)$$

This mechanism learns to assign higher weights to the most informative facial regions, such as the eyes and mouth. While reducing emphasis on less relevant areas. As a result, WGAP not only improves the aggregation of discriminative features but also yields interpretable heatmaps (see Fig. 4) that confirm the model's focus on key expression cues.

Throughout training, the model adaptively learns these weights jointly with the rest of the model, emphasizing the most discriminative regions while diminishing the significance of less relevant areas. This enables the model to concentrate on essential expressive signals, enhancing resilience to identity variations and background noise. Despite its benefits, WGAP introduces only a modest increase in parameters and computation, making it suitable for real-time FER applications.

### C. ENHANCED CONDITIONAL POSITIONAL ENCODING (ECPE)

Positional encoding is essential for transformers to maintain spatial structure; however, conventional techniques such as absolute positional encoding (APE) fail to accommodate

task-specific spatial relationships. Conditional positional encoding (CPE) presents a local context-aware method, yet it primarily addresses short-range dependencies.

In CPE, a generator is applied over each local patch of  $\mathbf{y}_0^m \in \mathbb{R}^{H_m \times W_m \times D}$  (where  $\mathbf{y}_0^m$  is defined in Table 1 and  $H_m$ ,  $W_m$  and  $D$  denote height, width, and channels) to yield the conditional positional encodings  $\mathbf{CPE}(\mathbf{y}_0^m) \in \mathbb{R}^{H_m \times W_m \times D}$ . The positional encoding generator is realized as a 2D convolution with kernel size  $k \geq 3$  and  $\frac{k-1}{2}$  zero paddings.

To overcome CPE's locality bias, we propose Enhanced CPE (ECPE), which fuses both local and global spatial awareness. We first apply the standard positional encoding generator (CPE( $\cdot$ )) to patches of  $\mathbf{y}_0^m$  in early stages, then incorporate a global encoding over the entire feature map by expanding kernel size in the final stage.

In contrast to conventional CPE, which emphasizes only local neighborhoods, ECPE integrates global context into the positional encoding, enabling the model to discern subtle expression patterns across various facial regions.

### D. OVERALL MODEL REPRESENTATION

These improvements are incorporated into the final architecture through a structured pipeline:

- The input image is partitioned into patches and embedded into feature vectors.
- The transformer encoder employs self-attention and feed-forward layers to process tokens.
- ECPE is implemented to encode both local and global spatial dependencies.
- WGAP is applied to the final spatial embeddings, and its output is concatenated with the class token in the two-stream aggregation step.
- The classification head receives the final feature vector and predicts the facial expression.

This procedure is outlined in Algorithm 1, with all operators and symbols defined in Table 1.

The proposed transformer-based framework effectively addresses the existing limitations in modeling subtle spatial relationships and global contextual information that are crucial for accurate facial expression recognition by incorporating the two-stream aggregation approach, weighted



**Algorithm 1** Enhanced Vision Transformer**Input:** Image  $\mathbf{x}$  of shape  $(H, W, C)$ **Parameters:**Patch size:  $P$ Number of patches:  $L = \frac{H \times W}{P^2}$ Embedding dimension:  $D$ Transformer blocks per stage:  $M$ Number of stages:  $M$ Output classes:  $K$ **Patch Embedding and Class Token Concatenation** $\mathbf{x}_{\text{patches}} \leftarrow \text{extract\_patches}(\mathbf{x}, P) \{ \mathbf{x}_{\text{patches}} \in \mathbb{R}^{L \times (P^2 C)} \}$  $\mathbf{x}_{\text{embed}} \leftarrow \text{PE}(\mathbf{x}_{\text{patches}})$  $\mathbf{x}_{\text{class}} \leftarrow \text{LearnableClassToken}() \in \mathbb{R}^{1 \times D}$ **Main Stages****for**  $m = 1$  to  $M$  **do**  **if**  $m == 1$  **then**     $\mathbf{z}_0^m \leftarrow [\mathbf{x}_{\text{class}}; \mathbf{x}_{\text{embed}}]$   **else**     $\mathbf{z}_0^m \leftarrow \text{PE}(\mathbf{y}_N^{m-1})$   **end if**   $\mathbf{y}_0^m \leftarrow \text{TE}(\mathbf{z}_0^m)$   **if**  $m == M$  **then**     $\mathbf{y}_1^m \leftarrow \text{TE}(\text{ECPE}(\mathbf{y}_0^m) + \mathbf{y}_0^m)$   **else**     $\mathbf{y}_1^m \leftarrow \text{TE}(\text{CPE}(\mathbf{y}_0^m) + \mathbf{y}_0^m)$   **end if**  **for**  $n = 2$  to  $N$  **do**     $\mathbf{y}_n^m \leftarrow \text{TE}(\mathbf{y}_{n-1}^m)$   **end for****end for****Classification Head** $\mathbf{y}_{\text{class}} \leftarrow \text{ExtractClassToken}(\mathbf{y}^M) \{ \in \mathbb{R}^{1 \times D} \}$  $\mathbf{y}_{\text{WGAP}} \leftarrow \text{WGAP}(\mathbf{y}_N^M)$  $\mathbf{y}_{\text{final}} \leftarrow \text{MLP}(\text{Concat}(\mathbf{y}_{\text{class}}, \mathbf{y}_{\text{WGAP}})) \{ \in \mathbb{R}^{1 \times K} \}$ **Output:** Class probabilities  $\mathbf{y}_{\text{final}}$  for each class

global average pooling, and enhanced conditional positional encoding. Collectively, these improvements establish a representation that is both discriminative and robust, making it appropriate for the capture of fine-grained facial expression details.

**IV. EXPERIMENTS**

The datasets and implementation details of our model are initially described in this section, and subsequently, extensive evaluations take place. The aim of our experiments is to evaluate the effectiveness of our proposed enhancements, which include enhanced conditional positional encoding (ECPE) and weighted global average pooling (WGAP), in the context of facial expression recognition.

**A. DATASETS**

Our proposed method is evaluated on three facial expression recognition datasets that are frequently employed:

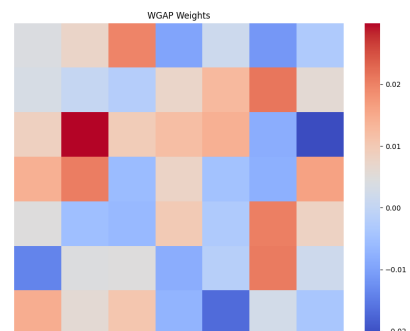
- **FER2013** [11]: This dataset contains 35,888 grayscale images of size  $48 \times 48$  pixels. The images are categorized into seven fundamental emotions: anger, disgust, surprise, happiness, sadness, and neutral. The standard split comprises 28,709 training samples and 3,589 test samples.
- **FERPlus** [2]: An extension of FER2013, FERPlus offers improved labels that are derived from crowd-sourced annotations. Each of the 35,888 images is relabeled into eight emotion classes, which includes contempt. This dataset is particularly beneficial for assessing the capacity to identify subtle expression differences.
- **RAF-DB** [23]: A dataset of 29,672 images that were gathered from the internet and represent real-world facial expressions. The dataset encompasses significant variations in pose, illumination, age, gender, and ethnicity. The single-label subset utilized in our experiments comprises 12,271 training and 3,068 test samples, which are categorized into seven fundamental emotion categories.

**B. IMPLEMENTATION DETAILS**

We utilize PyTorch [33] to implement our models, capitalizing on efficient GPU computation. Input images are resized to  $224 \times 224$  pixels, and data augmentation techniques, including random horizontal flips and color jittering, are employed. Models are initialized with weights pretrained on ImageNet and optimized using AdamW [26] with an initial learning rate of  $5 \times 10^{-4}$ , which is reduced by a factor of 10 after 30 epochs. Training is conducted for 100 epochs with a batch size of 32.

**C. ABLATION STUDY**

We conducted comprehensive ablation studies on the FER2013, RAF-DB, and FERPlus datasets to systematically evaluate the effectiveness of each architectural enhancement. The comparative performance of the different model configurations is presented in Table 2.

**FIGURE 4.** Visualization of learned WGAP weights.

**TABLE 2.** Ablation study of proposed enhancements on FER2013, RAF-DB, and FERPlus datasets. (Best-performing models in each dataset are highlighted in bold, and the best-performing models within each size category are underlined.)

| Model                          | Fer2013 (ACC%) | RAFDB (ACC%) | FerPlus(ACC%) | Params (M) | Flops (G) |
|--------------------------------|----------------|--------------|---------------|------------|-----------|
| Transformer + GAP (Small)      | 70.1           | 88.00        | 84.74         | 23.55      | 2.82      |
| Transformer + WGAP             | <u>71.5</u>    | <b>88.62</b> | <u>85.93</u>  | 23.55      | 2.82      |
| Transformer + WGAP + CT        | 70.4           | 88.38        | 84.94         | 23.56      | 2.83      |
| Transformer + WGAP + CT + ECPE | 70.5           | 88.49        | 85.31         | 23.57      | 2.83      |
| Transformer + GAP (Base)       | 70.8           | 88.17        | 86.70         | 55.31      | 8.36      |
| Transformer + WGAP             | 71.5           | 88.33        | 87.93         | 55.31      | 8.36      |
| Transformer + WGAP + CT        | 72.0           | <u>88.53</u> | <u>89.26</u>  | 55.32      | 8.37      |
| Transformer + WGAP + CT + ECPE | <u>72.1</u>    | 88.14        | 89.14         | 55.33      | 8.37      |
| Transformer + GAP (Large)      | 71.9           | 87.19        | 87.77         | 98.25      | 14.84     |
| Transformer + WGAP             | 71.8           | 87.81        | 88.42         | 98.25      | 14.84     |
| Transformer + WGAP + CT        | <b>72.2</b>    | 87.58        | 88.27         | 98.26      | 14.86     |
| Transformer + WGAP + CT + ECPE | 71.8           | <u>88.14</u> | <b>89.71</b>  | 98.28      | 14.86     |

Weighted Global Average Pooling (WGAP) consistently enhances accuracy in comparison to the standard Transformer with GAP, which is especially advantageous for smaller models. The finding implies that WGAP compensates for the limited capabilities of smaller models by directing them to concentrate on critical areas. These learned weight distributions are illustrated in Figure 4, which clearly delineates the most discriminative facial areas for facial expression recognition. As shown in Figure 4, central facial regions are assigned substantially higher weights than the corners, which exhibit near-zero weights, underscoring their minimal contribution to expression discrimination. In addition, the performance of all datasets is further improved by the integration of WGAP with the class token (CT), which emphasizes the complementary roles of global context and local spatial information in the capture of subtle facial expression details. The impact of Enhanced Conditional Positional Encoding (ECPE) is more configuration-dependent, providing significant improvements primarily in larger model variants.

Two configurations are chosen based on these results, as they offer the most robust overall performance in terms of both model complexity and accuracy:

- **Base model (Transformer + WGAP + CT):** Using only 55.33M parameters (8.37G FLOPs, 132 ms inference time, 1445 MB memory usage), this configuration achieves balanced performance across datasets, with accuracies of 72.0% on FER2013, 88.53% on RAF-DB, and 89.26% on FERPlus. It is particularly well-suited for scenarios that are practical and resource-constrained due to its efficiency.
- **Large model (Transformer + WGAP + CT + ECPE):** This model maintains competitive accuracy on FER2013 (71.8%) and exhibits exceptional performance on the RAF-DB (88.14%) and FERPlus (89.71%) benchmarks. This configuration is optimal for applications that prioritize accuracy and have fewer computational constraints, as it contains 98.28M parameters (14.86G FLOPs, 175 ms inference time, 2564 MB memory usage).

The experiments that were conducted have substantiated our design decisions and have demonstrated the effectiveness

of these specific architectural improvements in resolving the current challenges associated with facial expression recognition.

#### D. COMPARISON WITH STATE-OF-THE-ART

In this section, we benchmark the best-performing configurations of our proposed models against recent state-of-the-art methods. This comparison demonstrates the practical effectiveness of our improvements in the context of existing FER research.

##### 1) COMPARISON TO STATE-OF-THE-ART ON FER2013

We conduct a comparison between our best-performing model and the most recent state-of-the-art methods on FER2013. The results are presented in Table 3.

Our proposed method demonstrates strong performance, outperforming several state-of-the-art approaches while maintaining a relatively efficient model size. Although it does not achieve the highest accuracy, it provides a balanced trade-off between computational efficiency and accuracy, rendering it a viable alternative for real-world FER applications.

**TABLE 3.** FER2013 facial expression recognition accuracy for proposed model vs. state-of-the-art methods.(Best: Bold, Second Best: Underlined).

| Model                 | Accuracy (%) | Params(M) | Flops(G) |
|-----------------------|--------------|-----------|----------|
| Deep Emotion [28]     | 70.02        | -         | -        |
| Mini-ResEmoteNet [30] | 70.20        | 5.02      | -        |
| ViT+LMIM [6]          | 70.44        | -         | -        |
| CNN-HO [39]           | <u>72.16</u> | -         | -        |
| Ad-corre [8]          | 72.03        | 24.54     | 4.56     |
| LMHC [34]             | <b>74.42</b> | 32.4      | -        |
| Ours(base)            | 72.0         | 55.32     | 8.37     |
| Ours(large)           | 71.8         | 98.28     | 14.86    |

##### 2) COMPARISON TO STATE-OF-THE-ART ON RAF-DB

The accuracy of our model in comparison to state-of-the-art methods on RAF-DB is illustrated in Table 4.

Although our model does not surpass the most effective methods, such as POSTER, it achieves competitive accuracy on RAF-DB, surpassing several robust baselines, including DACL, Face2Nodes and C-EXPR-NET. This illustrates the

**TABLE 4. Comparison of proposed model to state-of-the-art methods on RAF-DB.(Best: Bold, Second Best: Underlined).**

| Model                 | Accuracy (%) | Params(M) | Flops(G) |
|-----------------------|--------------|-----------|----------|
| DACL [9]              | 80.44        | 103       | -        |
| Face2Nodes [15]       | 84.77        | 36.58     | 3.06     |
| Mini-ResEmoteNet [30] | 85.00        | 20.07     | -        |
| C-EXPR-NET [19]       | 87.50        | -         | -        |
| ViT + NAF [10]        | 88.14        | 51.8      | -        |
| POSTER [50]*          | <b>92.05</b> | 71.8      | 15.7     |
| Ours(base)            | <u>88.53</u> | 55.32     | 8.37     |
| Ours(large)           | 88.14        | 98.28     | 14.86    |

\* Model is pretrained on the MS-Celeb-1M dataset.

effectiveness of our proposed architecture enhancements in improving facial expression recognition, particularly in difficult real-world scenarios. A viable alternative for resource-constrained applications, our method also maintains a favorable balance between accuracy and model complexity.

### 3) COMPARISON TO STATE-OF-THE-ART ON FERPLUS

Table 5 demonstrates that our model surpasses numerous state-of-the-art methods in terms of competitive accuracy on FERPlus. Although some methods, such as POSTER, which benefits from additional pretraining on the Ms-Celeb-1M dataset, achieve greater accuracy, our model maintains a robust equilibrium between architectural efficiency and performance.

**TABLE 5. Comparison of our proposed model to state-of-the-art methods on FERPlus.(Best: Bold, Second Best: Underlined).**

| Model                   | Accuracy (%) | Params(M) | Flops(G) |
|-------------------------|--------------|-----------|----------|
| Improved MobileViT [14] | 80.30        | 0.95      | 0.27     |
| ESR [37]                | 87.15        | 20        | -        |
| ViT + NAF [10]          | 88.69        | 51.8      | -        |
| CSGResNet [16]          | 88.94        | 44        | -        |
| AMP-Net [24]            | 89.37        | 105.67    | -        |
| EAC [48]                | 89.64        | -         | -        |
| Face2Nodes [15]         | 89.71        | -         | -        |
| POSTER [50]*            | <b>91.62</b> | 71.8      | 15.7     |
| Ours(base)              | 89.26        | 55.32     | 8.37     |
| Ours(large)             | <u>89.71</u> | 98.28     | 14.86    |

\* Model is pretrained on the MS-Celeb-1M dataset.

The effectiveness of our proposed enhancements is confirmed by the experimental results from FER2013, RAF-DB,

and FERPlus. Ablation studies have consistently demonstrated the advantages of WGAP and ECPE in enhancing facial expression recognition. By effectively capturing global and local spatial contexts, the integration of the class token with global average pooling (GAP) further reinforces feature discrimination. Our model demonstrates clear advantages in performance and robustness, achieving superior or comparable accuracy in comparison to recent state-of-the-art methods. It is important to note that our proposed architecture provides a significantly smaller and faster model, which is more practical and efficient for real-world applications, despite the fact that POSTER, pre-trained on the MS-Celeb-1M dataset, achieves slightly higher accuracy.

### E. QUALITATIVE RESULTS

The model's performance is further analyzed by employing confusion matrices, t-SNE embeddings, and attention maps to visualize the framework's effectiveness. These visualizations offer a glimpse into the manner in which our model enhances the discriminability between facial expression classes and acquires spatially aware features.

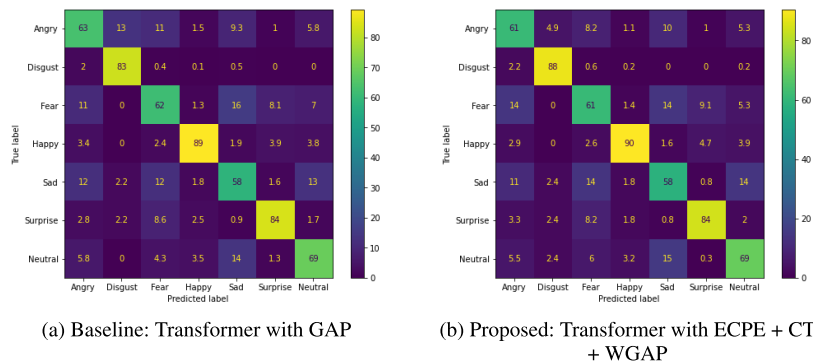
#### 1) CONFUSION MATRICES

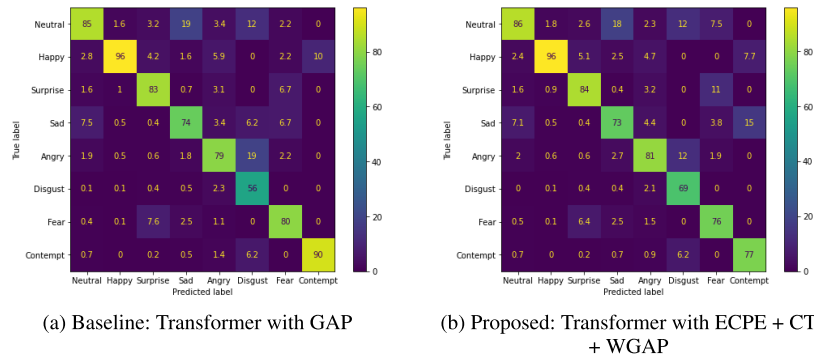
The baseline transformer model and our enhanced architecture are directly compared using confusion matrices. The baseline model (Transformer with class token + GAP) demonstrates a greater degree of confusion between similar facial expressions, such as fear and surprise, or anger and disgust, as illustrated in Figures 5 and 6.

On the contrary, our proposed model presents a substantial decrease in misclassifications. This implies that the network is able to more effectively capture subtle expression differences by focusing on relevant facial regions and incorporating richer spatial information as a result of the inclusion of Enhanced Conditional Positional Encoding (ECPE) and Weighted Global Average Pooling (WGAP).

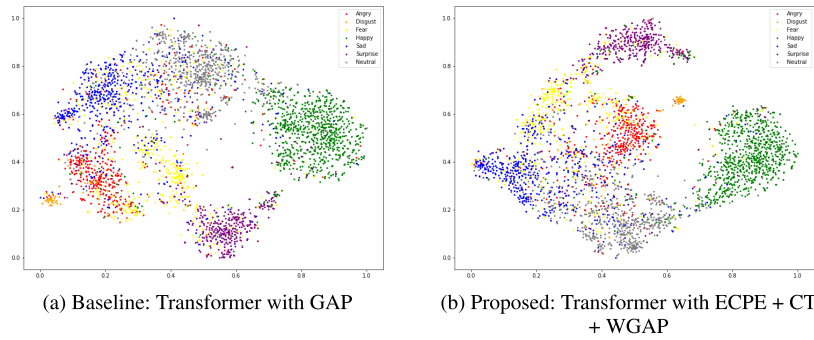
#### 2) t-SNE AND UMAP EMBEDDINGS

In order to more effectively demonstrate the discriminative capabilities of the acquired facial expression representations, we implemented dimensionality reduction techniques.

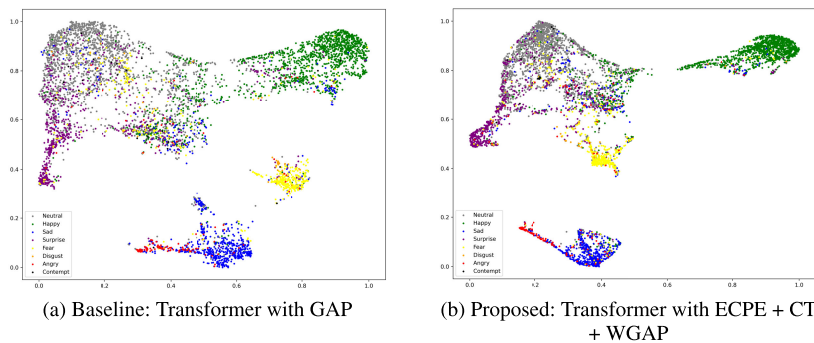
**FIGURE 5. Confusion matrices comparing baseline and proposed model on the FER2013 test set.**



**FIGURE 6.** Confusion matrices comparing baseline and proposed model on the FERPlus test set.



**FIGURE 7.** t-SNE visualization of expression embeddings from baseline and proposed models on the FER2013 test set.



**FIGURE 8.** UMAP visualization of expression embeddings from baseline and proposed models on the FERPlus test set.

In particular, we implemented Uniform Manifold Approximation and Projection (UMAP) on FERPlus and t-distributed Stochastic Neighbor Embedding (t-SNE) on embeddings from FER2013.

Figure 7 illustrates the t-SNE embeddings for FER2013, and Figure 8 shows the UMAP embeddings for FERPlus. Our proposed framework generates more distinct clusters than the baseline, particularly in expression pairs that pose a challenge. The UMAP visualization for FERPlus further emphasizes our model's improved capacity to differentiate subtle differences between similar facial expressions by generating denser clusters. This improved separation is attributed to WGAP, which ensures that critical facial regions

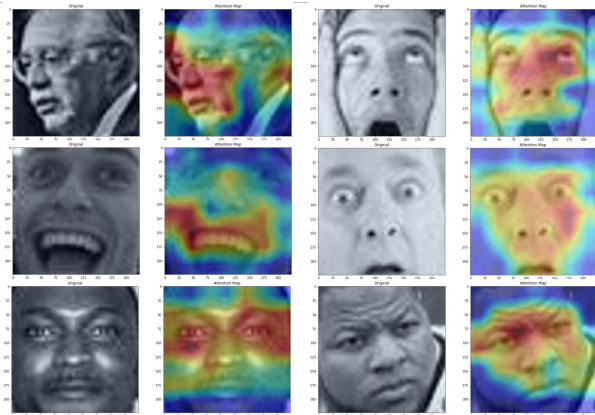
contribute more effectively to feature representations, and ECPE, which enhances spatial encoding by capturing both local and global relationships.

### 3) ATTENTION MAPS

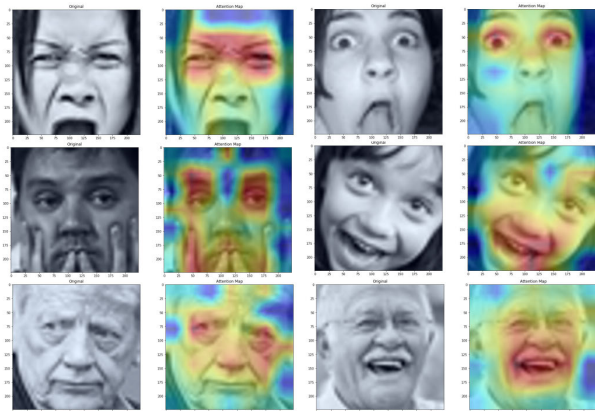
To interpret the model's focus when making predictions, we visualize the attention maps generated by our transformer network. Figures 9 and 10 illustrate sample attention heatmaps of our proposed architecture.

Our proposed model exhibits strong and focused attention on key facial areas, such as the eyes, eyebrows, and mouth. Specifically, in happy samples the attention concentrates on the mouth region, whereas in angry samples, it emphasizes





**FIGURE 9.** Attention map visualizations from the proposed model on sample images from FER2013.



**FIGURE 10.** Sample attention maps generated by the proposed model on images from the FERPlus dataset.

the eyes. However, we also observe occasional misallocation, for example, attention highlighting fingers, which may stem from the prevalence of hand-to-face poses in surprise and fear images. It can lead to errors when a happy image includes a hand gesture, underscoring limitations of the dataset. Such focused attention is a direct result of WGAP, which prioritizes critical regions for classification, and ECPE, which enhances spatial awareness by encoding global dependencies. These improvements allow the network to better distinguish fine-grained variations in facial expressions.

## V. CONCLUSION

This work introduces a transformer-based framework specifically designed for FER, incorporating architectural improvements such as WGAP and ECPE that enhance feature representation and discrimination. The two-stream transformer architecture improves the model's capacity to capture subtle spatial cues essential for distinguishing facial expressions. Competitive accuracy and efficiency in evaluations substantiate the effectiveness and adaptability of the proposed improvements.

Although transformer-based FER has been advanced by our research, further investigation is needed. Future work includes expanding the framework to additional datasets, integrating the introduced components into other enhanced transformer architectures, and exploring real-world applications such as human–computer interaction, driver monitoring, and mental health assessment.

Beyond FER, the modular design of our framework makes it applicable to other fine-grained classification tasks where subtle feature differences are crucial, establishing a foundation for future innovations in vision transformers across domains requiring precise and context-aware feature extraction.

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